

Study Group Finder

1. Business Requirements Document (BRD)

Project Name: Study Group Finder

Objective: Provide students with a web-based platform to create, discover, and join study groups based on subjects or courses. The platform simplifies collaboration by allowing scheduling, member management, and meeting location/link sharing.

Functional Requirements:

- **User Authentication:** Students can securely register and log in.
- **Create Study Group:** Users can create study groups with subject, description, date, time, member limit, and meeting location/link.
- **Join/Leave Groups:** Students can join available groups or leave anytime.
- **Search Groups:** Search by subject, course, or group name.
- **Member Management:** Prevent users from joining when the maximum member limit is reached.
- **Schedule & Meeting Details:** Display study session schedule and offline location or online meeting link.

Non-Functional Requirements:

- **Security:** Passwords must be encrypted using BCrypt.
- **Performance:** Search results should load within 2 seconds.
- **Responsiveness:** Fully functional on desktop and mobile devices.
- **Scalability:** Support multiple study groups simultaneously.

2. Technical Design Document (TDD)

A. Tech Stack

Frontend: React (Vite), Tailwind CSS

Backend: Node.js with Express

Database: PostgreSQL with Prisma ORM

Authentication: JWT

B. API Design

Method	Endpoint	Description
POST	/api/auth/register	Register a new student
POST	/api/auth/login	Authenticate and return JWT
GET	/api/groups	Fetch all study groups
POST	/api/groups	Create a study group
POST	/api/groups/:id/join	Join a study group
DELETE	/api/groups/:id/leave	Leave a study group
PUT	/api/groups/:id	Update study group
DELETE	/api/groups/:id	Delete study group

C. Database Schema (Prisma - Concept)

```

model User {
  id Int @id @default(autoincrement())
  name String
  email String @unique
  password String
  groups StudyGroup[]
}

model StudyGroup {
  id Int @id @default(autoincrement())
  groupName String
  subject String
  date DateTime
  time String
  location String?
  meetingLink String?
  maxMembers Int
  createdBy Int
}

```

D. Implementation Strategy

- Phase 1 (Database): Configure PostgreSQL and Prisma schema.
- Phase 2 (Backend): Build authentication and study group APIs.
- Phase 3 (Frontend): Develop dashboards, group creation, search, and join screens.
- Phase 4 (Deployment): Deploy frontend on Vercel and backend on Render/Railway.

How to Start

1. Create a shared GitHub repository.
2. Add BRD.md and TDD.md in the docs folder.
3. Review the documentation before writing code.

Sprint Planning

Sprint 1

Goal: Infrastructure & Authentication

- Database setup
- User registration
- User login

Sprint 2

Goal: Core Features

- Create study group
- Search groups
- Join/Leave groups

Sprint 3

Goal: Enhancements & QA

- Member limit management
- Schedule/location display

- Testing & deployment